

A REPORT

ON

COMPARATIVE ANALYSIS OF VECTOR-FREE RETRIEVAL STRATEGIES

FOR RAG: HYBRID SEARCH AND GRAPH-BASED APPROACHES

BY

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BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

(August, 2026)

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Abstract: Retrieval-Augmented Generation (RAG) systems depend on the quality of their retrieval stage, yet most published guidance derives from web-scale benchmarks whose properties differ from specialised corpora. This dissertation compares five retrieval strategies, BM25, normalised-cosine FAISS embeddings, an entity co-occurrence graph, hybrid Reciprocal Rank Fusion, and LLM reranking, on 130 Indian government policy documents segmented into 954 section-aware chunks. Five hypotheses were preregistered and evaluated on a 40-query locked test split, then on an independent 12-query holdout under frozen configurations, with a separate 150-answer panel for the faithfulness question. BM25 led the locked test (MRR@10 0.6071) and the holdout (0.6667), and hybrid fusion matched its recall without exceeding

its ranking quality. Graph retrieval gave the strongest directional evidence of any hypothesis, leading entity-relation and multi-hop categories, though two questions per category prevent confirmatory inference. Faithfulness could not be estimated: 300 of 301 verifiable claims were fully supported and the observed standard deviation of 0.0111 falls far below the preregistered threshold, leaving over-cautious abstention rather than hallucination as the dominant failure mode. No hypothesis reaches confirmatory status under the frozen statistical rules.

Signature of Student

Signature of Supervisor

Date:

Date:

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1. INTRODUCTION

1.1 Background and Motivation

Retrieval-Augmented Generation (RAG) has become the default architecture for grounding large language model outputs in external evidence. Rather than relying on parameters alone, a RAG system first retrieves passages relevant to the user query and then conditions generation on that evidence. Lewis et al. formalised the approach and demonstrated its value on knowledge-intensive tasks [1], and subsequent work has shown that retrieval augmentation measurably reduces hallucination in dialogue and question answering [15], [17]. Enterprises now deploy RAG for customer support, internal knowledge search, and citizen-facing information services, which makes the quality of the retrieval stage a matter of operational consequence rather than academic preference.

There is, however, a persistent gap between the retrieval literature and the corpora on which practitioners actually deploy. Most published comparisons of retrieval methods rest on web-scale benchmarks such as MS MARCO or Natural Questions, where dense neural retrievers trained on millions of query-passage pairs enjoy a well-documented advantage over lexical methods [5], [9]. Zero-shot evaluations across heterogeneous domains tell a more cautious story: Thakur et al. found that BM25 remains a hard baseline once retrievers leave their training distribution [11]. The question of which method to choose for a small, terminology-dense, domain-specific corpus is therefore far less settled than the headline benchmark results suggest.

Indian government policy documents are a demanding example of such a corpus. Scheme documentation is written in formal administrative language, saturated with exact names and acronyms such as PM-KISAN, PMMY, PMJAY, and SSY, and organised around explicit relationships between schemes, ministries, implementing agencies, and beneficiary groups. Citizens and service-desk operators query this

material with mixtures of exact scheme names, colloquial paraphrases, and multi-step eligibility questions. A retrieval failure here is not an abstract metric loss: it can mean a wrong answer about who qualifies for an income-support transfer or a health-insurance benefit.

The motivation for this dissertation is to replace assumption with measurement for exactly this setting. Instead of importing conclusions from web benchmarks, the work builds a controlled testbed of 22 public policy documents, defines five competing retrieval strategies spanning lexical, dense, graph-based, fusion, and LLM-reranked designs, and evaluates all of them under a preregistered protocol with an independent holdout set. The emphasis on vector-free strategies (BM25, entity graphs, and their fusion) reflects a practical concern: many organisations want strong retrieval without the operational cost of embedding pipelines and vector databases.

1.2 Problem Statement

Existing retrieval evaluations concentrate on large collections with hundreds of thousands of queries, using web search as the proxy task. Conclusions drawn there, most prominently that dense embeddings outperform BM25, do not transfer automatically to a domain corpus with bounded vocabulary, exact scheme nomenclature, and limited paraphrase variation [5], [11]. Whether semantic similarity computed from general-purpose embeddings adds anything over careful lexical matching on such a corpus is an open empirical question.

Equally unresolved is the cost side. LLM-based reranking can sharpen relevance judgments [7], but it introduces per-query inference cost and latency measured in seconds. A production service answering thousands of queries per day must know whether mechanical fusion of cheap retrievers captures most of that benefit. Finally, RAG systems are usually evaluated end to end, which conflates retrieval quality with generation behaviour; whether faithful generation actually depends on retrieval strength deserves a direct test.

The problem addressed by this dissertation is therefore: for a finite corpus of government policy documents, which retrieval strategy best balances ranking quality, robustness to unseen queries, and practical cost, and how do retrieval quality and generation faithfulness interact? The problem is answered through preregistered hypothesis testing on a pilot benchmark followed by frozen-configuration validation on an independent holdout set.

1.3 Research Objectives and Hypotheses

The work is organised around five preregistered hypotheses. Each was frozen, together with its test statistic and decision rule, before any evaluation was run.

H1. BM25 performs competitively with FAISS dense retrieval on exact-match and terminology-sensitive queries in the government policy domain.

H2. FAISS outperforms BM25 on paraphrased queries where the query vocabulary differs from the document vocabulary.

H3. An entity co-occurrence graph retriever outperforms both BM25 and FAISS on entity-relation and multi-hop queries.

H4. Hybrid retrieval combining BM25 and the graph retriever through Reciprocal Rank Fusion achieves the highest aggregate Mean Reciprocal Rank across all query categories.

H5. Generation faithfulness is independent of retrieval quality: strong and weak retrievers produce equally faithful answers under identical generation instructions.

Hypotheses H1 through H4 partition the retrieval design space; H5 tests the common assumption that the retrieval and generation stages of a RAG pipeline can be optimised separately. Supporting objectives include constructing a reproducible benchmark with human-validated relevance judgments, implementing all five retrievers in production-quality code with frozen configurations, and quantifying pilot-to-holdout generalisation.

1.4 Scope and Limitations

The evaluation proceeds on three frozen evidence tiers. The R4 benchmark corpus contains 130 policy documents segmented into 954 section-aware chunks, with 100 owner-reviewed questions split by category stratification into a 60-query development set used for all parameter selection and a 40-query locked test evaluated exactly once. An independent 12-query holdout, two questions in each of six categories, was authored afterwards and evaluated with every configuration frozen; the canonical hypothesis verdicts rest on this holdout. A separate Phase 9 panel of 150 generated answers across 30 questions and five systems supports the faithfulness question. An earlier 22-document, 34-query pilot exists in the project history and is retained only as development provenance; no result in this report is drawn from it.

Five retrieval systems are in scope: BM25 through the rank-bm25 library; normalised-cosine FAISS over a pinned all-MiniLM-L6-v2 revision; an entity co-occurrence graph built with NetworkX over all 954 chunks (3,781 nodes, 26,430 edges); hybrid fusion via Reciprocal Rank Fusion, evaluated both as an equal BM25-plus-graph combination and as a development-selected weighted BM25-FAISS-graph combination; and Prompt-RAG, which reranks candidates with Claude Haiku 4.5. Generation quality for H5 uses a fixed model and fixed system instructions across all retrievers.

Several directions are deliberately out of scope: fine-tuning domain-specific embedding models, cross-lingual retrieval over Indian-language versions of the documents, conversational multi-turn context, learned-to-rank fusion, and corpus scaling beyond the 22-document testbed. These are noted as future work in Section 6. Statistical power is the binding limitation of this study. The holdout provides two questions per category, and every paired confidence interval on the primary metric crosses zero. The protocol therefore reports effect directions with bootstrap intervals and treats no hypothesis as confirmed; Section 6.3 develops this constraint in full.

1.5 Organization of the Report

Section 2 reviews the literature on RAG, dense retrieval, lexical retrieval, graph-based retrieval, fusion and reranking methods, and retrieval evaluation methodology. Section 3 describes the corpus, the benchmark and holdout design, all five retriever implementations with their frozen configurations, the evaluation metrics, and the preregistered testing protocol. Section 4 presents the system architecture: the layered platform design, the ingestion and indexing pipelines, the retrieval and fusion subsystems, the evaluation harness, and the technology stack. Section 5 reports results for the five hypotheses, presenting pilot results, holdout results, and cross-dataset analysis for each. Section 6 discusses the findings, their implications for practitioners, and threats to validity. Section 7 states conclusions and recommendations. References follow in the WILP citation format, then a glossary of technical terms. Appendix A contains the complete benchmark tables, Appendix B the prompt templates, Appendix C the frozen configuration files, and Appendix D worked benchmark examples.

2. LITERATURE REVIEW

2.1 Retrieval-Augmented Generation: An Overview

Lewis et al. introduced RAG as a general framework in which a retriever selects k passages for a query and a sequence-to-sequence generator conditions on both the query and the retrieved evidence [1]. The framework builds on the contextual representation advances that BERT introduced [21] and on the observation, formalised in REALM, that a language model can be pre-trained jointly with a retriever so that evidence selection becomes part of the learned objective [22]. The two-stage decomposition allows each stage to be optimised independently, and it converts an open-ended generation problem into one that is at least partially auditable: the evidence passed to the generator can be inspected. Izacard and Grave showed that generative readers can aggregate evidence across many retrieved passages, strengthening the case for optimising retrieval recall [12].

The empirical case for retrieval augmentation is strongest on hallucination. Shuster et al. measured substantial reductions in fabricated content when dialogue models were augmented with retrieval [15], and the broader survey literature identifies missing or misleading evidence as a primary driver of hallucination in knowledge-grounded generation [17]. At the same time, retrieval imposes a ceiling: if no relevant passage appears in the retrieved set, the generator cannot recover the missing information, and conflicting retrieved passages can degrade answers even when a correct one is present [9], [12].

Gao et al. organise the modern RAG landscape into naive, advanced, and modular variants, and observe that most production deployments still rely on comparatively simple retrievers behind heavily engineered prompts [9]. Later designs add control flow around retrieval: Self-RAG trains a model to decide when to retrieve and to critique its own output, and to abstain when evidence is inadequate [29], a mechanism closely related to the abstention behaviour examined in this study under H5. That observation frames the present work: before investing in sophistication, a practitioner should know how much the simple

components actually deliver on their corpus. This dissertation holds generation fixed and studies the retrieval stage in isolation, which the H5 analysis then justifies empirically.

2.2 Dense Retrieval Methods

Dense retrieval encodes queries and documents as fixed-dimensional vectors and ranks by vector similarity. Karpukhin et al. demonstrated with DPR that a dual-encoder trained on question-passage pairs outperforms BM25 on open-domain question answering by wide margins [5], a result made possible by the contextual representations that BERT introduced [21]. Sentence-BERT made general-purpose sentence embeddings practical by training siamese BERT networks, so that similarity can be computed with inexpensive vector operations [3]. Training strategy proved as important as architecture: ANCE showed that mining hard negatives from the index itself substantially improves dense retriever quality [23]. Late-interaction designs such as ColBERT keep token-level representations and improve effectiveness at higher index cost [10]. On the systems side, FAISS provides exact and approximate nearest-neighbour search at scale [4].

The caveat that matters for this dissertation is distribution shift. The BEIR benchmark showed that dense retrievers trained on one distribution often fall behind BM25 when evaluated zero-shot on unfamiliar domains [11], a finding measured against training corpora such as MS MARCO [27] and Natural Questions [28] whose query distributions differ sharply from administrative text. Embeddings encode the semantic priors of their training corpora, mostly web text, news, and encyclopaedic prose. Administrative policy language, with its scheme names, statutory phrases, and acronym conventions, sits outside that distribution. The present work uses all-MiniLM-L6-v2, a widely deployed 384-dimensional model, deliberately without fine-tuning, so that the evaluation reflects what a practitioner gets off the shelf.

2.3 Sparse and Lexical Retrieval Methods

BM25 remains the reference lexical ranking function four decades into the probabilistic retrieval tradition. Robertson and Zaragoza give the standard derivation: a query-document score built from term frequency with saturation (parameter k_1) and document-length normalisation (parameter b) over inverse document frequency weights [2]. The weighting scheme rests on Sparck Jones's account of term specificity [20], and the parameter values used here are those established in the Okapi TREC-3 experiments [19]. Manning, Raghavan and Schuetze cover the surrounding machinery of tokenisation, indexing, and evaluation that makes lexical retrieval fast and predictable in practice [18].

The strengths of BM25 are exactly the properties that matter on specialised corpora: no training data requirement, exact matching of rare terms, interpretability of scores, and robustness to domain shift because the method carries no learned priors [2], [11]. Its weakness is the vocabulary mismatch problem; BM25 cannot bridge a query and a document that describe the same concept with disjoint words. Two families of methods attack this directly while keeping a sparse index: document expansion predicts likely queries and appends them to the passage before indexing [25], and learned sparse models such as SPLADE assign weights to expansion terms in the vocabulary space [24]. Neither is evaluated here, since the object of study is the unmodified lexical baseline. Whether that weakness is material depends on how much genuine paraphrase variation the query workload contains, which is an empirical property of the domain rather than of the algorithm. The tokenisation choices made for the BM25 implementation in this work, in particular the decision not to stem and not to remove stop words, are documented in Section 3.2 and turn out to be consequential.

2.4 Graph-Based Retrieval

Graph-based retrieval represents corpus structure explicitly: entities become nodes, relationships or co-occurrences become edges, and retrieval traverses the graph from query entities to related content.

Velickovic et al. demonstrated that attention over graph neighbourhoods captures relational structure that flat representations miss [8], within the broader programme of knowledge graph construction and use surveyed by Hogan et al. [32]. More recently, Edge et al. proposed GraphRAG, which builds an LLM-derived entity graph over a corpus and uses community summaries to answer global questions, reporting gains on query-focused summarisation [16].

The dependency that the graph literature consistently understates is extraction quality. A graph retriever can only traverse from entities it recognises in the query; every named-entity-recognition miss silently degrades the system to its fallback behaviour. On entity-dense scientific text this dependency is mild. On procedural policy language, where queries mix scheme names with colloquial descriptions, extraction coverage becomes the binding constraint. Neural sequence labelling has been the standard answer to extraction quality since Lample et al. [31], but it requires domain-annotated training data that policy corpora rarely provide. The entity co-occurrence graph built for this dissertation (Section 3.2) instruments this dependency directly by logging query-time seeding rates per category, which later proves decisive for interpreting H3.

2.5 Hybrid Fusion and LLM Reranking

Combining retrievers is an old and effective idea. Reciprocal Rank Fusion assigns each document a score equal to the sum of $1/(k + \text{rank})$ across input rankings and re-sorts; Cormack et al. showed that this parameter-light scheme outperforms Condorcet fusion and several learned methods [6]. RRF needs no score calibration between systems, which makes it attractive when fusing rankers whose score scales are incomparable, such as BM25 and a graph traversal count.

Neural reranking applies a stronger model to reorder a candidate list produced by a cheap first stage. Nogueira and Cho established the pattern with BERT-based passage reranking, roughly halving the error of the first-stage ranking on MS MARCO [7]; Lin et al. survey the design space that followed [26].

Instruction-tuned models extended the pattern to zero-shot listwise reranking, which Sun et al. found competitive with supervised rankers [30]. Instruction-following LLMs extend this to zero-shot relevance scoring: the model receives the query and each candidate and emits a graded relevance judgment. The trade-off is cost and latency, since every query now involves LLM inference over dozens of candidates. The Prompt-RAG system in this dissertation follows this design with Claude Haiku 4.5 scoring fifty BM25 candidates on an integer scale from zero to three, under a JSON schema that forces a complete, machine-checkable response (Section 3.2). Comparing it with mechanical RRF fusion on the same corpus gives a direct measurement of what LLM judgment adds over fusion, which is the substance of H4.

2.6 Evaluation Methodology for Retrieval Systems

The evaluation design of this dissertation borrows three practices from the information retrieval measurement tradition. The first is pooled relevance judging: because exhaustively judging every query-document pair is infeasible, the top results of all participating systems are pooled and judged, a method standardised by the TREC evaluations [13] and analysed in depth by Sanderson [33]. Pooling introduces a known bias against systems that retrieve relevant documents no pooled system found, which is acceptable here because all five systems under study contributed to the pool. The second is the choice of rank-sensitive metrics. Mean Reciprocal Rank rewards placing the first relevant result early, which mirrors how RAG consumes retrieval output, since generation attends most to the top of the context; nDCG generalises this to graded relevance with logarithmic discounting [14]. The third is preregistration, imported from the experimental sciences: fixing hypotheses, slices, statistics, and decision rules before measurement removes the temptation to select favourable analyses after the fact. Preregistration remains rare in retrieval benchmarking, and its use here, together with a frozen-configuration holdout, is a deliberate methodological contribution of the work. Small-sample honesty completes the picture: with category slices of four to seven queries, paired randomisation tests and bootstrap intervals are reported

with effect directions, and no claim rests on a single p-value. The choice of randomisation over parametric testing follows Smucker et al., who found permutation and randomisation tests better suited to retrieval evaluation than the t-test or sign test alone [34].

3. METHODOLOGY

3.1 Corpus and Benchmark Design

Corpus. The R4 benchmark corpus contains 130 Indian government policy documents drawn from public portals, including Pradhan Mantri Mudra Yojana (PMMY, micro-lending), Pradhan Mantri Kisan Samman Nidhi (PM-KISAN, farmer income support), Sukanya Samriddhi Yojana (SSY), Atal Pension Yojana (APY), Pradhan Mantri Jan Arogya Yojana (PMJAY), and further central and state schemes. A corpus audit before rebuilding confirmed zero exact duplicate documents, identified one near-duplicate pair and ten noisy-chunk candidates, and removed no review-sensitive document. All documents are in the public domain; no proprietary HCLTech data was used, consistent with the signed No Objection Certificate.

Chunking. The corpus was rebuilt into 954 balanced, section-aware chunks using 450-word windows with 60-word overlap. Section-aware segmentation respects document structure rather than cutting at fixed offsets, and document titles and section headings are retained inside the retrieval text so that a chunk carries its own context. Each chunk holds a stable chunk identifier and document identifier in a JSONL manifest, making every retrieval result traceable to its source. The average is about 7.3 chunks per document. This rebuild replaced an earlier 856-chunk segmentation; the change from fixed-window to section-aware chunking alone improved locked-test BM25 by 0.0416 MRR@10.

Benchmark queries and splits. One hundred owner-reviewed questions span six categories chosen to exercise the hypotheses: exact match, terminology-heavy, paraphrase, entity relation, multi-hop, and synthesis. These were divided by category-stratified split into 60 development queries and a 40-query locked test. All parameter selection, including BM25 tuning and hybrid weighting, used development queries only; the locked test was evaluated once. Twenty deterministic cross-document synthesis candidates, each drawing on two source programmes and at least two evidence chunks, were held separately as automated candidates.

Relevance judgments. Following standard pooling practice from TREC evaluation [13], retrieval results were pooled per query and judged on a graded scale. All 140 gold judgments inherited from the previous corpus revision were mapped onto R4 chunks through a crosswalk that the project owner reviewed row by row, retaining one intentional grade-zero mapping. Table 3.1 summarises the corpus and benchmark statistics.

Table 3.1: Corpus and benchmark statistics

Property	R4 benchmark	Independent holdout
Documents	130	Frozen R4 corpus
Chunks	954 (section-aware)	Frozen R4 corpus
Chunk window	450 words	Same
Chunk overlap	60 words	Same
Questions	100 (60 dev / 40 locked)	12 (2 per category)
Query categories	6	6 (same taxonomy)
Relevance judgments	140 owner-reviewed crosswalk	21 owner-reviewed
Parameter tuning	Development split only	None (frozen)
Evaluation passes	Locked test evaluated once	Zero-retry contract

Independent holdout. Twelve fresh questions, two in each of six categories, were authored after all system configurations were frozen, and carry 21 owner-reviewed relevance judgments. A zero-retry contract governed the run: each system executed each query exactly once, with no re-runs, no configuration edits, and no post-hoc question changes. Uncertainty is quantified with 10,000-sample whole-query paired bootstrap at seed 42. The holdout is the canonical evidence base for the hypothesis verdicts in Section 5. BM25, FAISS cosine, Entity Graph v4, and Hybrid R4 completed frozen evaluation. Prompt-RAG completed only 1 of 12 requests under the zero-retry contract; a later 12-of-12 execution repeated an ambiguously dispatched request and is therefore excluded from canonical inference and admissible only as exploratory sensitivity evidence. This exclusion is disclosed rather than concealed, in line with the discrepancy-disclosure requirement of the WILP report guidelines.

3.2 Retriever Implementations

All five retrievers are implemented in Python under `src/retrievers/`, share a common `RetrievalResult` interface returning chunk identifier, score, and rank, and persist frozen configurations as JSON alongside their indexes. This subsection records the implementation decisions that matter for interpreting the results.

BM25 (`src/retrievers/bm25_retriever.py`, 196 lines). The implementation uses the `rank-bm25` library with `BM25Okapi`, whose `k1` and `b` parameters were tuned on the 60-question development split and then frozen before the locked test [2]. Tokenisation is deliberately minimal: lowercase, strip punctuation, split on whitespace. There is no stemming, no lemmatisation, and no stop-word removal. This is a domain-driven choice, not an omission. Policy text depends on exact forms: stemming would corrupt scheme tokens, and removing function words would break multi-word names such as Ministry of Finance. The index is built once over all 954 chunks and pickled with its configuration for reproducibility; retrieval scores the full corpus and returns the top-k by descending score with sequential ranks.

FAISS dense retriever (`src/retrievers/faiss_retriever.py`). Chunks are encoded with sentence-transformers `all-MiniLM-L6-v2` at a pinned model revision, a 384-dimensional encoder pre-trained on general web and encyclopaedic text [3]. Embeddings are L2-normalised and indexed with inner-product search, giving normalised cosine similarity, and the index performs exact rather than approximate search [4]. An earlier revision of the system used unnormalised L2 distance; the move to normalised cosine is recorded here because it changes the ranking behaviour and therefore the reported numbers. The model is used strictly off the shelf, with no domain fine-tuning, because H2 tests what generic embeddings deliver on out-of-distribution text. Queries are encoded with the same pinned model at search time.

Entity co-occurrence graph v3.2 (`src/retrievers/entity_graph_v3.py`, over 500 lines). The graph is built in three stages. First, entity extraction combines two sources: trusted structured metadata (document titles, ministry and department fields, which give complete coverage over documents) and pattern-based

extraction from text, using rules for paired long-form/acronym expressions such as a scheme name followed by its bracketed acronym, statutory phrases ending in Act or Policy with an optional year, and controlled uppercase acronym tokens. Extraction is restricted to eight entity types (scheme, scheme alias, ministry, department, implementing agency, beneficiary group, law or policy, delivery institution, controlled acronym) with a fixed priority order. Second, entity labels are normalised with Unicode NFKC folding, casefolding, punctuation-to-space mapping, and whitespace collapse, then deduplicated with alias merging under a canonical label. Third, a NetworkX co-occurrence graph is built over all 954 R4 chunks: entities appearing in the same chunk are connected bidirectionally. The rebuilt graph holds 3,781 nodes and 26,430 edges. Version 4 of the retriever adds alias expansion, seed filtering, a hub penalty that damps highly connected generic entities, dual-entity coverage requirements, and a lexical fallback path.

At query time the retriever extracts entities from the query with the same rules. If at least one entity is found, the graph is traversed up to two hops from the seed entities, and chunks containing any traversed entity are ranked by matched-entity count subject to the hub penalty and dual-entity coverage rules. If no entity is found, the query takes the lexical fallback path. The fallback is instrumented, so the proportion of queries served by graph traversal rather than by fallback is recoverable from the run artefacts. The v4 rebuild materially improved graph performance over its predecessor, and the resulting system produces the strongest category-level evidence of any hypothesis in this study, as Section 5.3 reports.

Hybrid RRF (src/retrievers/hybrid_rrf_v1.py, 119 lines). The fusion layer combines validated top-50 rankings with rank-only Reciprocal Rank Fusion [6]: each chunk scores the sum of $1/(60 + \text{rank})$ over the systems that returned it, with $k = 60$, input and output depth 50, and a deterministic tie-break by ascending chunk identifier. Two variants were built: an equal-weight BM25-plus-graph combination matching the original H4 specification, and a development-selected weighted BM25-FAISS-graph combination. The holdout Hybrid R4 system is the three-component variant, which means it does not correspond exactly to

the two-component system H4 originally specified; this implementation mismatch is disclosed in Section 5.4. The frozen configuration (`configs/hybrid_rrf_v1_frozen.json`) fixes every one of these values, and the implementation emits a per-chunk audit trace recording each component rank, each RRF contribution, and whether a tie-break was required, so any fused ranking can be reproduced by hand. Because two chunks holding mirrored ranks in the two systems receive identical scores, the ascending-identifier tie-break keeps the output deterministic.

Prompt-RAG (`src/retrievers/prompt_rag_claude_v2.py`, over 200 lines). The reranker sends the query and exactly fifty BM25 candidates to Claude Haiku 4.5 (model `claude-haiku-4-5-20251001`) at temperature zero, instructing it to act as a relevance assessor for government policy queries and to score each candidate from 0 (not relevant) to 3 (highly relevant). The response is constrained by a JSON schema keyed to the candidate chunk identifiers, and a contract layer rejects any response with missing keys, extra keys, or out-of-range scores. Candidates are then reordered by score descending with the same ascending-identifier tie-break. Because the candidate pool is BM25 top-50, Prompt-RAG measures pure relevance-judgment ability on a fixed pool; it cannot retrieve anything BM25 missed. Candidate pools were built as deterministic BM25-top-25 plus FAISS-top-25 unions, whose depth-25 recall of 0.880 exceeds the depth-10 union recall of 0.825. Measured latency over 100 locked-test queries was 16,820 ms mean, 15,804 ms median, and 24,254 ms at the 95th percentile, under a 2.5 dollar experiment cost cap.

Table 3.2 consolidates the frozen configuration of every system for reference.

Table 3.2: Frozen configuration summary for all five systems

System	Key parameters (frozen)	Source file
BM25	k1/b tuned on dev split, no stemming	<code>bm25_retriever.py</code> (196 lines)
FAISS	all-MiniLM-L6-v2 pinned, normalised cosine	<code>faiss_retriever.py</code>
Entity Graph v4	8 entity types, 2-hop, 3781 nodes, 26430 edges	<code>entity_graph_v3.py</code> (500+ lines)

System	Key parameters (frozen)	Source file
Hybrid R4	k=60, dev-selected weights, depth 50	hybrid_rrf_v1.py (119 lines)
Prompt-RAG	Haiku 4.5, temp 0, BM25+FAISS union pool	prompt_rag_claude_v2.py (200+ lines)

3.3 Evaluation Metrics

The preregistered primary metric is Mean Reciprocal Rank at cutoff 10 (MRR@10): for each query, the reciprocal of the rank of the first relevant chunk in the top ten, zero if none appears, averaged over queries [13]. Secondary metrics are nDCG@10, which discounts gains logarithmically by rank following Jarvelin and Kekalainen [14]; Recall@10, the fraction of gold chunks retrieved; Precision@10 and Precision@5; and Hit Rate at 5 and 10. Metrics are computed per query and averaged, and also broken down by query category, because four of the five hypotheses are category-specific. The metric implementations live in `src/evaluation/metrics.py` and are unit-tested against hand-computed examples.

For H5, faithfulness is measured at claim level rather than by a coarse answer-level score. The Phase 9 panel comprises 150 frozen answers spanning 30 questions and five retrieval systems. Two reviewers independently scored every row, each completing all 150; reviewer claim-count agreement was exact on 150 of 150 rows, so no adjudication was required. Each verifiable claim in a non-abstaining answer is labelled fully supported, partially supported, unsupported, or contradicted against the supplied evidence, and answers containing no verifiable claim are recorded as abstentions rather than scored. Row-level faithfulness is the claim-weighted support rate. The preregistration additionally requires a minimum spread in this outcome before any association test may be run, a gate whose consequences Section 5.5 reports.

3.4 Preregistered Hypotheses and Testing Protocol

Each hypothesis was preregistered with its query slice, test statistic, and decision rule before any system was evaluated, and all configurations were frozen before the holdout run. The statistical machinery lives

in `src/evaluation/statistics.py`, implementing percentile bootstrap intervals after Efron and Tibshirani [36], randomisation tests following Smucker et al. [34], and Holm's sequentially rejective correction for multiplicity [35].

H1 is tested on the combined exact-match and terminology-heavy slice with a paired percentile bootstrap on the BM25 minus FAISS MRR difference. The decision rule is an equivalence band: a 95 percent confidence interval lying entirely inside plus or minus 0.05 supports the hypothesis; an interval spanning the band is inconclusive; an interval outside rejects it. In the canonical holdout this slice contains only four questions, which is why the criterion cannot be established in either direction.

H2 is tested on the paraphrase slice with a one-tailed paired sign-randomisation test of FAISS exceeding BM25 by at least 0.1 MRR, at alpha 0.05.

H3 is tested on the entity-relation and multi-hop slices with per-slice sign-randomisation tests of the graph exceeding both BM25 and FAISS, Holm-corrected across the two slices.

H4 is tested on the aggregate of each evidence tier with pairwise comparisons of hybrid against BM25, FAISS, and the graph; support requires hybrid to lead on aggregate MRR@10. Note that the system actually evaluated on the holdout is the weighted three-component hybrid, whereas H4 was written for the two-component BM25-plus-graph system, a mismatch disclosed with the verdict.

H5 is tested with Spearman rank correlation between retrieval MRR@10 and claim-level faithfulness across the Phase 9 panel, using 10,000 whole-question bootstrap samples at seed 42. The preregistration imposes a range-restriction gate before the association rule may be applied: faithfulness must show a standard deviation of at least 0.10 across at least three distinct values. If that gate fails, H5 is declared not estimable, and statistical non-significance may not be reported as evidence for the hypothesis.

Three protocol properties deserve emphasis. First, all parameter selection used the 60-question development split, and the 40-question locked test was evaluated exactly once, so locked-test figures are

not the product of repeated tuning. Second, configurations were frozen before the holdout, so holdout numbers cannot have been tuned toward. Third, the category samples are small enough that every main paired interval crosses zero, which is why Section 5 reports effect directions with intervals, declines to declare any hypothesis confirmed, and treats disagreement between the two tiers as a signal about resolution rather than as a contradiction to be resolved in favour of the more convenient result.

4. SYSTEM ARCHITECTURE

This section describes the architecture of the experimental platform built for the study. Section 3 specified what each retriever does; this section explains how the components fit together, how data moves through the system, and which technologies implement each layer. The platform was designed around three requirements that follow directly from the research protocol: every run must be reproducible from a frozen configuration, every retrieval result must be traceable to a source chunk, and any retriever must be swappable without touching the evaluation harness.

4.1 Architectural Overview

The platform follows a four-layer design. The ingestion layer converts source policy documents into a normalised chunk manifest. The indexing layer builds one index per retrieval strategy from that shared manifest. The retrieval layer exposes all five strategies behind a single interface that returns ranked chunk identifiers with scores. The evaluation layer runs query sets against any subset of retrievers, computes metrics against relevance judgments, and writes immutable run artefacts. A generation layer sits alongside for the faithfulness study, consuming retrieved evidence under a fixed instruction template.

Layer separation is what makes the comparison fair. Because all five retrievers consume the identical chunk manifest and emit the identical result structure, no strategy gains an advantage from preprocessing differences or output formatting. The evaluation harness never learns which retriever produced a ranking; it receives a list of chunk identifiers and scores, and computes metrics identically for all of them.

[DIAGRAM PLACEHOLDER — Figure 4.1]

- *Four horizontal layers, top to bottom: Ingestion, Indexing, Retrieval, Evaluation.*
- *Ingestion box: source policy documents (PDF/HTML) to normalisation to chunker to chunks.jsonl manifest.*
- *Indexing box: five parallel branches from chunks.jsonl to BM25 pickle index, normalised-cosine FAISS index, NetworkX entity graph, and (no index) fusion and reranking layers.*
- *Retrieval box: unified RetrievalResult interface fed by all five strategies.*
- *Evaluation box: query set plus qrels to metrics engine to run artefacts (rankings, metrics, audit traces).*

- Side annotation: frozen configuration JSON files feeding the indexing and retrieval layers.
- Dashed side box: generation layer consuming top-k evidence under a fixed system prompt (used for H5 only).

Figure 4.1: End-to-end system architecture of the retrieval evaluation platform

4.2 Data Ingestion and Chunking Pipeline

Ingestion begins with 130 public policy documents retrieved from government portals. Each document is normalised to plain text with Unicode NFKC folding, whitespace collapse, and removal of navigation boilerplate, then assigned a stable document identifier together with structured metadata fields for title, ministry, and department. These metadata fields later become trusted entity sources for the graph retriever, which is why they are captured at ingestion rather than inferred later.

The chunker segments each normalised document into section-aware passages using 450-word windows with 60-word overlap, respecting document structure rather than cutting at fixed offsets, and retaining titles and section headings inside the retrieval text. Overlap prevents a fact that straddles a boundary from being lost to both chunks. Every chunk receives a stable identifier and a back-reference to its document, and the complete set of 954 chunks is written once to a JSONL manifest that all downstream indexers read. The manifest is content-addressed: any change to chunking produces a new manifest rather than mutating the existing one, so historical run artefacts remain interpretable.

[DIAGRAM PLACEHOLDER — Figure 4.2]

- Left: 130 source documents from MyScheme, PM-KISAN, PMJAY and allied portals.
- Normalisation stage: NFKC folding, whitespace collapse, boilerplate stripping.
- Metadata extraction stage: document id, title, ministry, department fields.
- Chunker stage: 450-word section-aware window, 60-word overlap, headings retained.
- Right: chunks.jsonl manifest, 954 section-aware chunks, average 7.3 chunks per document.
- Annotation showing overlap region shared between chunk n and chunk $n+1$.

Figure 4.2: Document ingestion and chunking pipeline

4.3 Retrieval Subsystem Design

Each retriever implements a common contract: build an index from the chunk manifest, persist that index with its configuration, and answer a query with a ranked list of results carrying chunk identifier, document identifier, text, score, and rank. Scores are strategy-specific and deliberately not normalised across strategies, because the fusion layer uses ranks rather than scores and the metrics depend only on ordering.

The lexical and dense branches are structurally simple: one index each, built once and cached. The graph branch is the most involved, and its internal structure explains the results reported later. Entity candidates are drawn from two sources of very different reliability, structured document metadata and pattern matches over body text, then normalised and merged into a registry of canonical entities with aliases. Co-occurrence within a chunk creates bidirectional edges, producing a graph of 3,781 nodes and 26,430 edges over all 954 chunks. At query time the same extraction rules run against the query; if they yield at least one entity, traversal proceeds up to two hops and candidate chunks are ranked by matched-entity count under a hub penalty and dual-entity coverage rules, and if they yield nothing, the query falls through to the lexical branch. That fallback path is instrumented, so the share of queries genuinely served by traversal is recoverable from run artefacts.

[DIAGRAM PLACEHOLDER — Figure 4.3]

- *Left half, offline construction: chunk manifest feeding two extraction paths, trusted structured metadata and regex pattern matching over text.*
- *Normalisation and alias merging into a canonical entity registry across eight entity types.*
- *Co-occurrence edge builder producing the NetworkX graph over 954 chunks, 3,781 nodes and 26,430 edges.*
- *Right half, query time: query text to the same extraction rules, then a decision diamond labelled entities found.*
- *Yes branch: two-hop traversal from seed entities to candidate entity set to chunk scoring by matched-entity count.*
- *No branch: lexical fallback retrieval, instrumented and logged per query.*
- *Annotation of v4 additions: alias expansion, seed filtering, hub penalty, dual-entity coverage.*

Figure 4.3: Entity graph construction and query-time traversal

4.4 Fusion and Reranking Layer

The fusion layer consumes validated top-50 rankings and produces a single ordering by Reciprocal Rank Fusion with k equal to 60, in two variants: an equal-weight BM25-plus-graph combination and a development-selected weighted BM25-FAISS-graph combination. Because RRF depends only on ranks, no score calibration between the two very different scoring schemes is required. The implementation emits an audit trace for every fused result recording each component rank, each reciprocal contribution, the fused total, and whether the deterministic ascending-identifier tie-break was invoked, which makes any fused ranking reconstructable by hand.

The reranking layer is architecturally different. It takes a deterministic BM25-top-25 plus FAISS-top-25 union as its candidate pool, sends the query and candidates to the language model in a single stateless request at temperature zero, and receives graded relevance scores under an enforced JSON schema keyed to the candidate identifiers. A contract layer validates the response before it is used: missing keys, extra keys, duplicate identifiers, or out-of-range scores all raise errors rather than silently degrading the ranking. This strictness matters because a partially malformed response would otherwise produce a plausible but unreproducible ordering.

[DIAGRAM PLACEHOLDER — Figure 4.4]

- *Upper path, hybrid fusion: BM25, FAISS and graph top-50 rankings entering an RRF combiner with k equal to 60.*
- *Combiner detail box showing the formula, one over sixty plus rank, summed across components.*
- *Tie-break block: equal scores resolved by ascending chunk identifier.*
- *Output: fused top-50 plus a per-chunk audit trace record.*
- *Lower path, reranking: BM25-top-25 plus FAISS-top-25 union pool into a request builder, then a single stateless model call.*
- *Response path: JSON schema enforcement, contract validation, integer scores zero to three, reorder by score descending.*
- *Latency annotations on each path: roughly 200 milliseconds for fusion versus 16.8 seconds mean measured for reranking.*

Figure 4.4: Hybrid fusion and LLM reranking data flow

4.5 Evaluation Harness and Experiment Control

The evaluation harness is the component that enforces the research protocol. It reads a query set, a retriever selection, and a relevance judgment file, executes each query against each selected retriever, and writes run artefacts containing the full rankings, per-query metrics, aggregate metrics, and the configuration hashes of every participating component. Metrics are computed by a single module shared across all systems and unit-tested against hand-worked examples.

Two control mechanisms deserve description because they are what make the holdout meaningful. Configuration freezing serialises every tunable parameter to JSON before a run and records its hash in the run artefact, so a later claim that a configuration was unchanged is verifiable rather than asserted. The zero-retry contract governs holdout execution: each system attempts each query exactly once, failures are recorded as failures, and no re-run is permitted. The Prompt-RAG holdout exclusion reported in Section 3.1, where only 1 of 12 requests completed, is a direct consequence of this contract operating as intended.

[DIAGRAM PLACEHOLDER — Figure 4.5]

- *Inputs on the left: query set (40-query locked test or 12-query holdout), retriever selection, relevance judgments, frozen configuration files.*
- *Central runner block executing the query-by-retriever matrix.*
- *Metrics engine block computing MRR@10, nDCG@10, Recall@10, Precision@5 and @10, hit rate.*
- *Statistics block: paired bootstrap, sign-randomisation, Holm correction, Spearman correlation.*
- *Outputs on the right: immutable run artefacts containing rankings, per-query metrics, aggregates, configuration hashes.*
- *Control annotations: configuration freeze gate before execution, zero-retry contract on the holdout path.*

Figure 4.5: Evaluation harness and experiment control flow

4.6 Generation Layer for the Faithfulness Study

The generation layer exists solely to support the H5 analysis and is held constant across retrievers. It receives the top-k evidence chunks from whichever retriever is under test, inserts them into a fixed instruction template that requires the model to answer only from the provided evidence and to abstain explicitly when the evidence is insufficient, and returns an answer together with the evidence actually

supplied. Two reviewers then score every output at claim level, labelling each verifiable claim as fully supported, partially supported, unsupported, or contradicted, with abstentions recorded separately so that declining to answer is never scored as unfaithful. One caveat applies: the Phase 9 panel spans two prompt-contract strata, 50 legacy records and 100 V3 records, so an identical-prompt claim across all 150 answers is not available and is disclosed in Section 5.5.

[DIAGRAM PLACEHOLDER — Figure 4.6]

- *Retriever under test supplying top-k evidence chunks into a prompt assembler.*
- *Fixed instruction template block: answer only from provided evidence, abstain if insufficient.*
- *Generation model block with fixed decoding parameters, identical across all five retrieval conditions.*
- *Two output branches: claim-bearing answer (81 of 150), or explicit abstention (69 of 150).*
- *Two-reviewer claim-level scoring block: each claim fully supported, partially supported, unsupported, or contradicted.*
- *Analysis block: range-restriction gate, then Spearman correlation between retrieval MRR and faithfulness.*

Figure 4.6: Generation layer and faithfulness evaluation path

4.7 Technology Stack

Table 4.1 records the technology selected at each layer together with the reason for the choice. Selections favour transparency and reproducibility over raw performance, which is appropriate for a 954-chunk research corpus; Section 7 notes where these choices would change at production scale.

Table 4.1: Technology stack by architectural layer

Layer	Technology	Version / Config	Rationale
Language	Python	3.11	Ecosystem coverage for IR and NLP tooling
Ingestion	Custom normaliser	450w/60w section-aware	Structure-respecting chunks; +0.0416 MRR@10 measured
Lexical index	rank-bm25 (BM25Okapi)	k1=1.5, b=0.75	Pure Python, no Java dependency, fully transparent
Embeddings	sentence-transformers	all-MiniLM-L6-v2 (pinned)	Off-the-shelf baseline at a pinned revision; no fine-tuning
Vector index	FAISS	Normalised cosine	Exact search removes

Layer	Technology	Version / Config	Rationale
		(exact)	approximation as a confound
Graph store	NetworkX	DiGraph, 3781 nodes	In-memory, inspectable, adequate at corpus scale
Entity extraction	Regex + structured metadata	8 entity types	Deterministic and auditable; no opaque NER model
Fusion	Custom RRF	k=60, depth 50, weighted	Rank-only; weights selected on development split
Reranking	Claude Haiku 4.5	temp 0, JSON schema	Graded relevance judgment; 16.8 s mean measured latency
Numerics	NumPy	1.26	Vectorised scoring and metric computation
Statistics	Custom + SciPy	Bootstrap, randomisation	Preregistered tests implemented explicitly
Testing	pytest	Unit + contract tests	Metric correctness verified against worked examples
Artefacts	JSON / JSONL	Immutable per run	Reproducibility and post-hoc auditability
Version control	Git	Tagged phase commits	Configuration freezing tied to commit history

4.8 Deployment and Reproducibility Considerations

The platform runs as a set of command-line entry points on a single machine; no distributed infrastructure is required at this corpus size. Indexes are built once and cached, so a full five-system evaluation of the 40-query locked test completes in minutes apart from the reranking condition, whose measured mean latency of 16.8 seconds per query dominates total runtime. Reproducing any reported number requires the chunk manifest, the frozen configuration files, and the query set, all of which are versioned together.

Scaling this design to a production corpus would change three decisions. Exact cosine search would give way to an approximate index once the corpus outgrows memory-resident exhaustive scan, accepting a small recall loss for logarithmic query cost. The in-memory NetworkX graph would need a persistent graph store as node and edge counts grow. And the reranking layer would require caching and batching to

keep per-query cost acceptable at volume. None of these changes affect the comparative conclusions, because all three concern throughput rather than ranking quality.

[DIAGRAM PLACEHOLDER — Figure 4.7]

- *Single-host deployment boundary containing all components.*
- *Persistent artefacts on disk: chunks.jsonl, BM25 pickle index, FAISS index files, serialised entity graph, frozen configuration JSON.*
- *In-process components: retriever implementations, fusion layer, metrics engine, statistics module.*
- *External dependency outside the host boundary: language model API endpoint used by the reranking and generation layers.*
- *Annotations marking the three components that would change at production scale: approximate vector index, persistent graph store, reranking cache and batching layer.*

Figure 4.7: Component and deployment view

5. RESULTS

Results are reported on two frozen evidence tiers. The R4 locked test (40 questions, evaluated once after all tuning on the development split) establishes system-level performance on the full 130-document corpus. The independent 12-question holdout, run under a zero-retry contract, is the canonical basis for the hypothesis verdicts. Each hypothesis subsection reports locked-test evidence where available, holdout evidence, and a verdict phrased within the limits the sample size permits.

One constraint governs every claim in this section. All main paired confidence intervals on the primary metric cross zero. The evidence therefore supports descriptive and directional statements, not decisive claims of superiority, and no hypothesis in this study reaches confirmatory status under the frozen statistical rules.

Table 5.1: R4 locked-test retrieval results (40 questions, evaluated once)

System	MRR@10	nDCG@10	Recall@10
BM25 (section-aware)	0.6071	0.6460	0.8625
Hybrid R4 (weighted)	0.5938	0.6306	0.8625
Prompt-RAG Claude	0.5842	0.6469	0.8875
Entity Graph v4	0.4579	0.4724	0.6750
FAISS (normalised cosine)	0.4113	0.4451	0.6750

Two ablations frame these numbers. Section-aware chunking improved locked-test BM25 over the previous fixed-window corpus by 0.0416 MRR@10, 0.0875 Recall@10, and 0.0470 nDCG@10, so a substantial share of the leading system's advantage comes from corpus preparation rather than from the ranking function. Equal-weight BM25-plus-graph fusion scored 0.5350 MRR@10, below BM25 alone, showing that undifferentiated fusion dilutes a strong component; only the development-weighted three-system variant recovered BM25-level recall.

Table 5.2: Independent holdout results (12 questions, frozen configurations)

System	MRR@10	nDCG@10	Recall@10	Hit@10
BM25	0.6667	0.6938	0.8611	0.9167
Hybrid R4	0.6597	0.6855	0.8611	0.9167
FAISS cosine	0.5444	0.5685	0.7500	0.8333
Entity Graph v4	0.5162	0.4907	0.6528	0.7500
Prompt-RAG	Excluded	-	-	-

The BM25 holdout MRR@10 of 0.6667 carries a bootstrap 95 percent interval of [0.4583, 0.8750], which illustrates the resolution available at twelve questions. Prompt-RAG completed only 1 of 12 requests under the zero-retry contract; the later 12-of-12 execution repeated an ambiguously dispatched request and is excluded from canonical inference.

Table 5.3: Holdout MRR@10 by category (2 questions per category)

Category	BM25	FAISS	Graph v4	Hybrid
Exact match	1.0000	1.0000	0.7500	1.0000
Terminology-heavy	0.2500	0.4167	0.5556	0.2500
Paraphrase	0.6667	0.5000	0.0000	0.6250
Entity relation	0.5000	0.7500	1.0000	0.7500
Multi-hop	0.5833	0.2500	0.6667	0.5833
Synthesis	1.0000	0.3500	0.1250	0.7500

5.1 Hypothesis 1: BM25 Competitive with FAISS on Exact-Match and Terminology Queries

Locked-test evidence. BM25 leads the locked test at 0.6071 MRR@10 against 0.4113 for FAISS, and leads on nDCG@10 and Recall@10 as well. At category level the two systems tie on exact-match hit rate, while BM25 leads terminology-heavy MRR@10 at 0.3708 against 0.0179.

Holdout evidence. The holdout splits the hypothesis in two. Exact-match MRR@10 is tied at 1.0000 for both BM25 and FAISS, so the exact-match half of H1 holds cleanly. The terminology-heavy half does not: BM25 scores 0.2500 against FAISS 0.4167, with Entity Graph v4 highest at 0.5556. BM25 is therefore

not the best system on the terminology slice in the canonical evidence, which is the opposite of the locked-test direction.

Verdict: partial support; formal equivalence inconclusive. The preregistered rule required a paired confidence interval lying wholly inside plus or minus 0.05. Only four combined exact-match and terminology questions exist in the holdout, so that criterion cannot be established either way. The defensible claim is narrower than the hypothesis as written: BM25 remains highly effective for exact-match retrieval, tying the dense system at a perfect score, but the evidence does not show BM25 as the best system across the broader terminology-heavy slice.

Interpretation. The disagreement between the two tiers is itself informative. On 40 locked-test questions BM25 dominates terminology; on 2 holdout terminology questions it does not. With two questions per category, a single unlucky question moves the category mean by half its range, and no reading of these numbers should treat the holdout category figures as stable estimates. The honest summary is that BM25 is a strong general baseline whose advantage on terminology-heavy language is supported at 40 questions and not reproduced at 2.

5.2 Hypothesis 2: FAISS Outperforms BM25 on Paraphrase Queries

Locked-test evidence. Paraphrase MRR@10 is 0.2125 for BM25 against 0.3063 for FAISS, the one category where the dense system leads on the locked test. Both scores are low in absolute terms; paraphrase is the hardest category for every system evaluated.

Holdout evidence. The direction reverses: BM25 scores 0.6667 against FAISS 0.5000 on the two holdout paraphrase questions.

Verdict: not supported by current holdout; construct validity limited. The canonical evidence favours BM25, contrary to the hypothesis. The more important finding concerns the test itself rather than the outcome. Several questions in the paraphrase slice retain exact document terminology, so they do not

create the lexical mismatch the category was designed to produce. A test that fails to instantiate its own construct cannot cleanly reject the underlying claim about semantic retrieval.

Interpretation. H2 was not supported in this benchmark, and that statement should not be generalised into a claim that dense retrieval fails on paraphrase in this domain. The locked test, with 8 paraphrase questions rather than 2, actually favours FAISS. What the study establishes is that its own paraphrase slice was insufficiently adversarial. A separately preregistered, paraphrase-only follow-up with questions written to avoid document vocabulary is required before H2 can be reconsidered.

5.3 Hypothesis 3: Graph Retrieval Outperforms on Entity-Relation and Multi-Hop Queries

Holdout evidence. Entity Graph v4 leads entity-relation MRR@10 at 1.0000 against BM25 0.5000 and FAISS 0.7500. It also leads multi-hop at 0.6667 against BM25 0.5833 and FAISS 0.2500. These are the two categories the hypothesis named in advance, and the graph leads both.

Locked-test evidence. The aggregate picture is less favourable: Graph v4 sits at 0.4579 MRR@10 overall, below BM25 and the hybrid systems. Category-level locked-test figures show the graph competitive on multi-hop at 0.6438 but behind BM25 on entity relation. The v4 rebuild, with alias expansion, hub penalty, seed filtering, and dual-entity coverage over the 954-chunk corpus, substantially improved on the previous graph revision while remaining below BM25 in aggregate.

Verdict: strongest directional support among the five hypotheses; not statistically confirmed. Each holdout category contains only two questions, which prevents reliable confirmatory inference. The result must be phrased as point-estimate evidence rather than proof.

Interpretation. H3 is the one hypothesis whose preregistered prediction was borne out in the direction predicted, on the categories named in advance, in the canonical evidence tier. That is a meaningful outcome and it is also a fragile one: perfect entity-relation performance on two questions is exactly the kind of figure that regresses under replication. The graph also scored 0.0000 on holdout paraphrase and

0.1250 on synthesis, so its strength is genuinely category-specific rather than general. The appropriate claim is that structured retrieval showed the most promising relational behaviour observed in this study and warrants a larger confirmatory evaluation, not that graph retrieval has been shown superior.

5.4 Hypothesis 4: Hybrid Fusion Achieves Highest Aggregate MRR

Holdout evidence. BM25 aggregate MRR@10 of 0.6667 exceeds Hybrid R4 at 0.6597, and BM25 also leads nDCG@10 at 0.6938 against 0.6855. Recall@10, Precision@10, and Hit@10 tie exactly at 0.8611, 0.1500, and 0.9167.

Locked-test evidence. The same direction holds: BM25 0.6071 against Hybrid R4 0.5938. The equal-weight BM25-plus-graph variant performed worse still at 0.5350, below BM25 by a clear margin.

Verdict: not supported. Fusion maintained BM25-level recall but did not deliver the highest aggregate MRR on either tier.

Interpretation and a disclosed mismatch. The holdout hybrid is the weighted three-component BM25-FAISS-graph system, whereas H4 as preregistered specified BM25 plus graph. The system tested is therefore not exactly the system hypothesised, and the verdict should be read with that mismatch in view. Setting it aside, the practical finding is consistent across tiers: fusion bought recall parity with BM25 and cost a small amount of ranking precision. The equal-weight result is the more instructive one, since it shows that fusing a strong component with a weaker one dilutes the strong component unless the weights are tuned. Prompt-RAG offers the counterpoint on the locked test, where it achieved the best Recall@10 at 0.8875 and the best nDCG@10 at 0.6469 while trailing BM25 on MRR@10, at a measured mean latency of 16.8 seconds per query against roughly 200 milliseconds for fusion. Neither fusion nor reranking displaced the lexical baseline on the primary metric.

5.5 Hypothesis 5: Retrieval Quality and Answer Faithfulness

Panel construction. The Phase 9 panel comprises 150 frozen answers across 30 questions and five retrieval systems. Two reviewers independently scored all 150 rows with exact claim-count agreement on every row, so no adjudication was required and no formula errors were found.

Table 5.4: Phase 9 claim-level faithfulness outcomes

Outcome	Count	Share
Total answers	150	100%
Claim-bearing answers	81	54.0%
Abstentions (no verifiable claim)	69	46.0%
Total verifiable claims	301	100%
Fully supported	300	99.7%
Partially supported	1	0.3%
Unsupported	0	0%
Contradicted	0	0%

Primary analysis. Claim-weighted faithfulness is 0.9983. Row-level faithfulness takes only two distinct values with a sample standard deviation of 0.0111. The Spearman correlation between MRR@10 and faithfulness is 0.1487, with a 95 percent bootstrap interval of [0.1189, 0.2563] over 10,000 whole-question samples at seed 42, of which 6,471 were non-constant and therefore usable.

Verdict: not estimable. The preregistration requires a faithfulness standard deviation of at least 0.10 across at least three distinct values before the association rule may be applied. The observed spread is 0.0111 across two values, so the gate fails and the hypothesis can receive neither confirmatory support nor a clean rejection from this panel. Under the frozen rules, statistical non-significance may not be reported as evidence in favour of H5.

Why the earlier framing was wrong. An interim analysis reported faithfulness as constant across retrievers and concluded that faithfulness is independent of retrieval quality because prompting enforces

it. That conclusion exceeded the evidence and is withdrawn. A ceiling effect and an independence result are observationally similar and statistically distinct: when an outcome has almost no variance, a null correlation is uninformative rather than meaningful. Nothing in this panel licenses a causal claim that prompt design rather than retrieval determines faithfulness. Testing that claim requires a design that deliberately induces variation in faithfulness, for example by degrading evidence quality or relaxing the abstention instruction, which no run in this project did.

Sensitivity. Treating partially supported claims as zero credit raises the standard deviation only to 0.0222 and leaves ρ unchanged at 0.1487, so the range criterion still fails. A legacy contract stratum yields 30 claim-bearing answers with standard deviation 0.0183 and ρ 0.2435; the V3 semantic contract yields 51 claim-bearing answers with faithfulness constant at 1.0, where ρ is undefined. Two prompt-contract strata exist across the panel, 50 legacy records and 100 V3 records, which prevents an identical-prompt claim across all 150 answers. This heterogeneity is disclosed rather than smoothed over.

What the panel does establish. Non-abstaining answers are almost perfectly grounded: zero of 301 scored claims are unsupported or contradicted. The dominant failure mode is over-cautious abstention rather than hallucination, with 69 of 150 outputs containing no verifiable claim. An exploratory follow-up (H5b) tested whether retrieval quality predicts answer production rather than faithfulness, and found no clear association, with all bootstrap intervals crossing zero: MRR against answered gives ρ -0.091 with interval [-0.330, 0.166], nDCG gives 0.062 with [-0.198, 0.323], and Recall gives 0.139 with [-0.140, 0.431]. The strong descriptive pattern was category dependence, with abstention rates of 90.0 percent on entity-relation, 83.3 percent on multi-hop, 43.3 percent on terminology, and 6.7 percent on both exact-match and paraphrase questions. Because H5b was formulated after the faithfulness ceiling was observed, it is exploratory and inconclusive.

6. DISCUSSION

6.1 Key Findings

Finding 1: no hypothesis reached confirmatory status. Across five preregistered hypotheses, two frozen evidence tiers, and a two-reviewer faithfulness panel, not one hypothesis satisfied its own decision rule. H1 obtained partial support with formal equivalence inconclusive, H2 and H4 were not supported, H3 obtained the strongest directional support without statistical confirmation, and H5 proved not estimable. This is the central result of the study and it should be reported as such rather than recast as a tally of wins and losses. Preregistration is what makes the outcome informative: because the decision rules were fixed before measurement, a null return is a genuine finding about the evidence rather than an absence of effort.

Finding 2: corpus preparation moved results more than algorithm choice. Rebuilding the corpus from fixed-window into 954 section-aware chunks, with headings retained in the retrieval text, improved locked-test BM25 by 0.0416 MRR@10, 0.0875 Recall@10, and 0.0470 nDCG@10, without any change to the ranking function. That improvement is comparable in size to the gaps separating whole families of retrieval methods elsewhere in these results. For practitioners deciding where to spend effort, this is the most transferable observation in the study.

Finding 3: structured retrieval showed the most promising relational behaviour. Entity Graph v4 led both categories H3 named in advance, entity relation at 1.0000 and multi-hop at 0.6667, in the canonical holdout. It also scored 0.0000 on paraphrase and 0.1250 on synthesis, so the strength is specific rather than general, and two questions per category cannot confirm it. The finding is a direction worth pursuing at larger scale, not a demonstrated advantage.

Finding 4: fusion bought recall parity, not ranking gains. Hybrid systems matched BM25 recall on both tiers while trailing it on MRR@10, and equal-weight fusion of BM25 with the weaker graph

component scored 0.5350 against BM25 alone at 0.6071. Fusion is a way to preserve coverage across heterogeneous components, not a reliable route to better top-rank precision. LLM reranking showed the complementary profile on the locked test, taking the best Recall@10 at 0.8875 and best nDCG@10 at 0.6469 while trailing on MRR@10, at a measured mean latency of 16.8 seconds per query.

Finding 5: the generation stage abstained rather than hallucinated. Of 301 verifiable claims, 300 were fully supported and none were unsupported or contradicted, while 69 of 150 outputs contained no verifiable claim at all. Abstention was strongly category dependent, at 90.0 percent for entity-relation and 6.7 percent for exact-match questions. The system's practical weakness is silence where an answer was possible, not fabrication. This finding is descriptive; the ceiling in the faithfulness outcome prevents it from supporting any claim about what causes faithfulness.

6.2 Implications for Practice

The first implication is procedural. A team choosing a retrieval strategy for a specialised corpus should expect that a study of this size will not settle the question, and should design its evaluation accordingly: preregister the decision rule, hold out a genuinely independent set, and be prepared to report that the intervals crossed zero. The alternative, selecting a winner from point estimates at small sample sizes, produces confident conclusions that do not survive replication.

The second is about where effort pays. Section-aware chunking with headings preserved delivered a measurable improvement at low cost and no inference-time expense. Before comparing ranking functions, it is worth confirming that the retrieval unit itself is well formed.

The third concerns fusion and reranking. Equal-weight fusion diluted a strong component; weighted fusion recovered recall but not ranking quality. Neither displaced a well-prepared lexical baseline on the primary metric. LLM reranking bought recall and nDCG at roughly two orders of magnitude more latency, which

is a defensible trade only where those metrics drive the application. Fusion remains attractive where component coverage differs, provided weights are selected on a development split rather than assumed. The fourth concerns generation. An abstention-permitting instruction produced near-perfect grounding and a 46 percent abstention rate. Teams adopting this pattern should monitor abstention as a first-class quality metric, because a system that declines to answer nearly half the time is failing users even though it never fabricates. Whether faithfulness would survive a weaker instruction is untested here and should not be assumed.

6.3 Threats to Validity

Statistical conclusion validity. This is the dominant threat. The canonical holdout contains twelve questions, two per category, and every main paired confidence interval on MRR@10 crosses zero. The BM25 aggregate interval alone spans [0.4583, 0.8750]. At two questions per category, one question shifts a category mean by half its range, which is why the terminology-heavy result inverts between the 40-question locked test and the 12-question holdout. Bootstrap resampling cannot manufacture resolution that the sample does not contain. Every category-level number in Section 5 should be read as a point estimate with wide and largely unquantified uncertainty, and no ordering among systems within a category is established.

Construct validity. Two constructs failed to instantiate. The paraphrase slice retains exact document terminology in several questions, so it does not create the lexical mismatch H2 was designed to test; the hypothesis was therefore evaluated against a weaker construct than intended. More seriously, the faithfulness outcome exhibits a ceiling: 300 of 301 claims fully supported, standard deviation 0.0111, two distinct values. An outcome with almost no variance cannot support a correlation analysis, which is why H5 is declared not estimable rather than confirmed or rejected. The earlier conclusion that faithfulness is

independent of retrieval because prompting enforces it is withdrawn, as a ceiling effect and genuine independence are observationally similar and statistically distinct.

Internal validity. Relevance judgments were pooled from the systems under study, a practice standardised by TREC but known to disadvantage systems outside the pool [13]; since all comparisons here are among pooled systems, the comparative conclusions are unaffected, though absolute recall figures are optimistic. The R4 qrels derive from a 140-row crosswalk mapping earlier judgments onto rebuilt chunks; the owner reviewed every row, but a crosswalk is a weaker instrument than fresh judging on the new segmentation, and minimum frozen-reference coverage of 0.5111 indicates that some mappings are partial. The H4 verdict carries a disclosed implementation mismatch: the holdout hybrid is a three-component weighted system whereas H4 specified two components. The Phase 9 panel spans two prompt-contract strata, 50 legacy and 100 V3 records, preventing an identical-prompt claim across all 150 answers. The faithfulness panel used two reviewers with exact claim-count agreement, which is a strength, but agreement on counts is not the same as agreement on graded support judgments.

External validity. The corpus is 130 documents in one administrative domain and one language, and 954 chunks remains small enough that term statistics behave differently than they would at production scale. Findings are claims about terminology-dense administrative corpora at modest scale, not about retrieval in general. Questions were authored and reviewed within the project rather than sampled from production logs, so their phrasing may resemble document vocabulary more closely than genuine citizen queries would, a confound that plausibly favours lexical matching. The holdout tests generalisation to unseen questions within the domain, not across domains, corpora, or languages.

What the evidence will and will not carry. The defensible claims are that BM25 is a strong baseline on this corpus, that section-aware chunking improved it measurably, that graph retrieval showed the most promising relational direction, that fusion preserved recall without improving ranking, and that

generated answers abstained rather than hallucinated. The claims this study cannot carry are that any system is superior to another, that dense retrieval fails on paraphrase in this domain, that graph retrieval outperforms lexical retrieval on relational queries, or that faithfulness is independent of retrieval quality. Section 7 is written to stay inside the first list.

7. CONCLUSIONS AND RECOMMENDATIONS

This dissertation compared five retrieval strategies for RAG over a corpus of 130 Indian government policy documents segmented into 954 section-aware chunks, using five preregistered hypotheses, a 40-question locked test, an independent 12-question holdout under a zero-retry contract, and a 150-answer two-reviewer faithfulness panel.

The principal conclusion is that no hypothesis reached confirmatory status under the frozen statistical rules. H1 received partial support, holding for exact-match retrieval where BM25 and FAISS tied at 1.0000 but not across the terminology-heavy slice, with formal equivalence inconclusive on four combined questions. H2 was not supported, as the holdout favoured BM25 on paraphrase, though the paraphrase construct was itself compromised by questions retaining document terminology. H3 obtained the strongest directional support of the five, leading both entity-relation and multi-hop categories, but two questions per category prevent confirmation. H4 was not supported, with BM25 exceeding hybrid fusion on aggregate MRR@10 across both tiers, subject to a disclosed component mismatch. H5 is not estimable: with 300 of 301 claims fully supported and a faithfulness standard deviation of 0.0111 against a preregistered threshold of 0.10, the range-restriction gate fails and no association can be tested.

This conclusion must not be restated as three hypotheses proved and two failed. Such wording would exceed the sample sizes, the confidence intervals, and the evaluation validity of the study. The contribution lies in the discipline of the negative result: preregistered rules, frozen configurations, a single evaluation of the locked test, an independent holdout, disclosed protocol violations, and a withdrawn overreaching claim about faithfulness. Two substantive positive findings survive: section-aware chunking improved locked-test BM25 by 0.0416 MRR@10 without changing the ranking function, and the generation stage produced no unsupported or contradicted claims across 301 scored claims, failing instead through a 46 percent abstention rate.

Recommendations. First, treat corpus preparation as a first-order design decision; the chunking rebuild delivered a larger measured gain than several algorithmic substitutions. Second, tune fusion weights on a development split, since equal-weight fusion measurably diluted the stronger component. Third, monitor abstention alongside faithfulness, because a system that never fabricates but declines half of all questions is not yet useful. Fourth, size evaluations to the claims intended; twelve questions cannot separate retrieval systems, and any team wanting confirmatory verdicts should plan for substantially larger category slices from the outset. Fifth, preregister decision rules and honour them, including when they return not estimable.

Future work. The most direct extension is a confirmatory evaluation of H3 with adequate category sample sizes, since it produced the strongest directional evidence here. A paraphrase-only follow-up with questions written to avoid document vocabulary would give H2 a valid construct. Testing whether faithfulness depends on retrieval requires a design that induces variance in the outcome, for example by degrading evidence quality or varying the abstention instruction, which no run in this project attempted. Further directions include domain fine-tuned embeddings, three-hop and weighted graph traversal, learned fusion weights, cross-lingual evaluation over Indian-language scheme documents, and scaling the corpus to observe how the lexical advantage evolves with vocabulary growth.

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GLOSSARY

Abstention: A generated response declining to answer because the retrieved evidence is insufficient. Recorded separately from faithfulness scoring; 69 of 150 panel outputs.

BM25: A probabilistic lexical ranking function scoring documents by term frequency, inverse document frequency, and length normalisation.

Chunk: A section-aware segment of a document (here a 450-word window with 60-word overlap) serving as the unit of retrieval.

Dense retrieval: Retrieval that ranks by similarity between learned vector representations of queries and documents.

Embedding: A fixed-dimensional vector representation of text produced by a neural encoder.

Entity seeding rate: The fraction of questions from which the entity extractor recognises at least one entity, enabling graph traversal rather than lexical fallback.

FAISS: A library for efficient exact and approximate nearest-neighbour search over dense vectors.

Faithfulness: The degree to which a generated answer makes only claims supported by the retrieved evidence. Measured here at claim level rather than per answer.

Hallucination: Generated content not supported by, or contradicting, the provided evidence.

Holdout set: Questions kept independent of all development and tuning, used only for final validation. Here, 12 questions forming the canonical verdict evidence.

Locked test: A held-back split evaluated exactly once after all tuning is complete. Here, 40 of the 100 R4 questions.

Not estimable: A preregistered verdict used when a gating condition fails, so a hypothesis can be neither supported nor rejected.

Range restriction: Insufficient variance in an outcome, which prevents meaningful correlation analysis. The cause of the H5 verdict here.

Ceiling effect: Scores clustered at the maximum, leaving no room to detect differences between conditions.

Section-aware chunking: Segmentation that respects document structure and retains headings, rather than cutting at fixed offsets.

Zero-retry contract: An execution rule permitting exactly one attempt per system per question, with failures recorded as failures.

MRR (Mean Reciprocal Rank): The average over queries of the reciprocal rank of the first relevant result.

nDCG: Normalised Discounted Cumulative Gain; a ranking metric that discounts relevance gains logarithmically by rank.

NER (Named Entity Recognition): Automatic identification of entity mentions, such as scheme or ministry names, in text.

Preregistration: Fixing hypotheses, test statistics, and decision rules before running an evaluation.

Prompt-RAG: The LLM reranking system in this study; Claude Haiku 4.5 scores BM25 candidates for relevance.

Qrels: Relevance judgments mapping query-document pairs to graded relevance labels.

RAG (Retrieval-Augmented Generation): An architecture that retrieves evidence for a query and conditions text generation on it.

Reranking: Reordering a candidate list from a first-stage retriever using a stronger scoring model.

RRF (Reciprocal Rank Fusion): A fusion method scoring each document by the sum of $1/(k + \text{rank})$ across input rankings.

Sparse retrieval: Retrieval over term-based representations, such as BM25 over an inverted index.

Tokenisation: Splitting text into units (tokens) prior to indexing or encoding.

APPENDIX A: DETAILED BENCHMARK TABLES

A.1 Benchmark Overview

This appendix reports complete retrieval metrics on both frozen evidence tiers. The R4 locked test comprises 40 category-stratified questions over 130 documents (954 section-aware chunks), evaluated once after all tuning on a separate 60-question development split. The independent holdout comprises 12 questions with 21 owner-reviewed judgments, evaluated under frozen configurations and a zero-retry contract, and is the canonical basis for the hypothesis verdicts. Prompt-RAG is excluded from canonical holdout analysis due to the dispatch protocol violation recorded in Section 3.1. All main paired confidence intervals on MRR@10 cross zero; these tables are descriptive.

A.2 System Configurations

R4 systems: BM25 (rank-bm25, k1 and b tuned on the development split, no stemming); FAISS (pinned all-MiniLM-L6-v2, normalised cosine, exact search); Entity Graph v4 (3,781 nodes, 26,430 edges over 954 chunks, 2-hop traversal with alias expansion, seed filtering, hub penalty, dual-entity coverage and lexical fallback); Hybrid R4 (RRF k = 60, depth 50, development-selected weights over BM25, FAISS and graph); Prompt-RAG (Claude Haiku 4.5, temperature 0, graded scoring over a BM25-top-25 plus FAISS-top-25 union pool). Holdout systems use identical frozen configurations.

A.3 Full Metric Breakdown: R4 Locked Test (40 questions)

Table A.1: R4 locked-test metrics, all five systems

System	MRR@10	nDCG@10	Recall@10	Prec@10	Hit@10
BM25	0.6071	0.6460	0.8625	0.1200	0.9000
Hybrid R4	0.5938	0.6306	0.8625	0.1200	0.9000
Prompt-RAG	0.5842	0.6469	0.8875	0.1225	0.9250
Graph v4	0.4579	0.4724	0.6750	0.0950	0.7500

System	MRR@10	nDCG@10	Recall@10	Prec@10	Hit@10
FAISS cosine	0.4113	0.4451	0.6750	0.0975	0.7500

Table A.2: R4 locked-test MRR@10 by category

Category	BM25	FAISS	Graph v4	Hybrid R4
Exact match	0.9167	0.3813	0.3125	0.7292
Terminology-heavy	0.3708	0.0179	0.3646	0.3875
Paraphrase	0.2125	0.3063	0.3125	0.2396
Entity relation	0.7917	0.7750	0.6563	0.8750
Multi-hop	0.7438	0.5764	0.6438	0.7375

Table A.3: Corpus rebuild ablation, locked test

Variant	MRR@10	Recall@10
R3 fixed-window BM25	0.5655	0.7625
R4 section-aware BM25	0.6071	0.8500
R3 FAISS baseline	0.4341	0.6250
R4 normalised-cosine FAISS	0.4113	0.6625
R4 equal BM25-Graph hybrid	0.5350	0.7750
R4 weighted 3-system hybrid	0.5938	0.8500

A.4 Independent Holdout (12 questions, canonical)

Table A.4: Holdout aggregate metrics with bootstrap intervals

System	MRR@10	MRR 95% CI	nDCG@10	Hit@10
BM25	0.6667	[0.4583, 0.8750]	0.6938	0.9167
Hybrid R4	0.6597	crosses zero vs BM25	0.6855	0.9167
FAISS cosine	0.5444	crosses zero vs BM25	0.5685	0.8333
Graph v4	0.5162	crosses zero vs BM25	0.4907	0.7500

Table A.5: Holdout MRR@10 by category (2 questions each)

Category	BM25	FAISS	Graph v4	Hybrid R4
Exact match	1.0000	1.0000	0.7500	1.0000

Category	BM25	FAISS	Graph v4	Hybrid R4
Terminology-heavy	0.2500	0.4167	0.5556	0.2500
Paraphrase	0.6667	0.5000	0.0000	0.6250
Entity relation	0.5000	0.7500	1.0000	0.7500
Multi-hop	0.5833	0.2500	0.6667	0.5833
Synthesis	1.0000	0.3500	0.1250	0.7500

Every paired confidence interval on MRR@10 crosses zero. At two questions per category, a single question shifts a category mean by half its range, which is why the terminology-heavy ordering differs between Table A.2 and Table A.5. No ordering within a category is established by this evidence.

A.5 Phase 9 Faithfulness Panel

Table A.6: Phase 9 panel composition and outcomes

Quantity	Value	Note
Frozen answers	150	30 questions x 5 systems
Reviewer A / B completion	150 / 150	Exact claim-count agreement
Adjudications required	0	All rows NOT_REQUIRED
Claim-bearing answers	81	54.0% of outputs
Abstentions	69	46.0% of outputs
Total verifiable claims	301	-
Fully supported	300	99.7%
Partially supported	1	0.3%
Unsupported / contradicted	0	0%
Claim-weighted faithfulness	0.9983	-
Faithfulness SD	0.0111	Two distinct values
Spearman rho (MRR)	0.1487	95% CI [0.1189, 0.2563]
Preregistered SD threshold	0.10	Not met; H5 not estimable

A.6 Statistical Test Summary

Table A.7: Preregistered hypothesis outcomes

Hyp.	Canonical evidence	Key figures	Verdict
H1	Holdout exact + terminology (n=4)	Exact tied 1.000; term. 0.250 vs 0.417	Partial support; equivalence inconclusive
H2	Holdout paraphrase (n=2)	BM25 0.667 vs FAISS 0.500	Not supported; construct limited
H3	Holdout entity-relation + multi-hop	Graph 1.000 and 0.667, leads both	Strongest directional support; unconfirmed
H4	Holdout and locked test aggregate	BM25 0.6667 vs Hybrid 0.6597	Not supported; component mismatch
H5	Phase 9 panel (150 answers)	SD 0.0111 vs 0.10 threshold	Not estimable (range restriction)

No hypothesis reaches confirmatory status. This table must not be summarised as a count of hypotheses proved and failed; the sample sizes and interval widths do not support that framing.

APPENDIX B: PROMPT TEMPLATES

This appendix reproduces the instruction templates used by the two components that call a language model. Both are held fixed for the duration of the study; any change would have invalidated the frozen-configuration guarantee described in Section 3.4.

B.1 Prompt-RAG Reranking System Instruction

The reranker receives this instruction as the system message, with the query and all fifty candidates supplied as a single JSON user message. Temperature is zero and the response format is constrained by a JSON schema keyed to the candidate identifiers.

```

You are a relevance assessor for government policy retrieval.
You will receive one question and exactly 50 candidate passages.
Score every candidate for its relevance to the question using this scale:
  3 = directly answers the question
  2 = relevant, contains part of the answer
  1 = related context, does not answer the question
  0 = not relevant
Rules:
- Score every candidate. Do not omit any chunk_id.
- Do not invent chunk_ids that were not provided.
- Judge only the text supplied. Do not use outside knowledge.
- Return integers only, in the required JSON object.
```

B.2 Response Schema Contract

The response is validated before use. Missing identifiers, unexpected identifiers, duplicates, non-integer values, and scores outside the range zero to three all raise a contract error and abort the query rather than producing a degraded ranking.

```
{
  "type": "json_schema",
  "schema": {
    "type": "object",
    "required": ["candidate_scores"],
    "additionalProperties": false,
    "properties": {
      "candidate_scores": {
        "type": "object",
        "required": [<all 50 candidate chunk_ids>],
        "additionalProperties": false,
        "patternProperties": {
          "^chunk-[0-9]+$": {"type": "integer", "minimum": 0, "maximum": 3}
        }
      }
    }
  }
}
```

B.3 Generation Instruction for the Faithfulness Study

This template is used identically across all five retrieval conditions in the H5 analysis. The explicit abstention clause is the mechanism that the results in Section 5.5 identify as the source of retrieval-independent faithfulness.

You answer questions about Indian government welfare schemes.

Use ONLY the evidence passages provided below. Do not use prior knowledge.
If the evidence does not contain enough information to answer, reply exactly:
"Cannot answer from provided evidence."
Do not speculate, do not generalise beyond the passages, and do not fill gaps.
Cite the chunk_id of each passage you rely on.

EVIDENCE:
{{retrieved_chunks}}

QUESTION:
{{question}}

B.4 Human Scoring Rubric

Table B.1: Claim-level faithfulness labels (Phase 9)

Label	Definition	Panel count
Fully supported	Every element of the claim appears in the evidence	300
Partially supported	Claim is partly evidenced, partly extrapolated	1
Unsupported	Claim has no basis in the supplied evidence	0
Contradicted	Claim conflicts with the supplied evidence	0

Scoring operates on individual claims rather than whole answers. Two reviewers labelled all 150 rows independently with exact claim-count agreement, so no adjudication was required. Answers containing no verifiable claim are recorded as abstentions (69 of 150) and are never scored as unfaithful, since declining to answer for lack of evidence is the behaviour the instruction requires. Row-level faithfulness is the claim-weighted support rate, which produced only two distinct values across the panel and a standard deviation of 0.0111.

APPENDIX C: FROZEN SYSTEM CONFIGURATIONS

Every configuration below was serialised to JSON and hash-recorded before the holdout evaluation, per the freezing procedure in Section 4.5. They are reproduced verbatim so that any reported result can be regenerated.

C.1 BM25 Configuration

```
{
  "system": "bm25",
  "library": "rank-bm25 (BM25Okapi)",
  "k1": 1.5,
  "b": 0.75,
  "tokenizer": "lowercase + punctuation_strip + whitespace_split",
  "tuning": "development_split_only",
  "stemming": false,
  "stopword_removal": false,
  "index_path": "indexes/bm25/bm25_index.pkl",
  "status": "frozen_before_results"
}
```

C.2 FAISS Configuration

```
{
  "system": "faiss",
  "model_name": "sentence-transformers/all-MiniLM-L6-v2",
  "embedding_dim": 384,
  "index_type": "IndexFlatIP (normalised cosine)",
  "similarity_metric": "cosine",
  "normalize_embeddings": true,
  "model_revision": "pinned",
  "fine_tuned": false,
  "status": "frozen_before_results"
}
```

C.3 Entity Graph Configuration

```
{
  "system": "entity_graph_v4",
  "graph_library": "networkx.DiGraph",
  "entity_types": ["scheme", "scheme_alias", "ministry", "department",
                  "implementing_agency", "beneficiary_group",
                  "law_or_policy", "delivery_institution",
                  "controlled_acronym"],
  "nodes": 3781,
  "edges": 26430,
  "max_hops": 2,
  "normalization": "NFKC + casefold + punct_to_space + collapse",
  "chunk_ranking": "matched_entity_count",
  "unseeded_query_policy": "fallback_to_bm25",
  "v4_additions": ["alias_expansion", "seed_filtering", "hub_penalty",
                  "dual_entity_coverage", "lexical_fallback"],
  "status": "frozen_before_results"
}
```

```
}
```

C.4 Hybrid RRF Configuration

```
{
  "system": "hybrid_bm25_graph_rrf",
  "components": ["bm25", "faiss_cosine", "entity_graph_v4"],
  "rrf_k": 60,
  "weights": "development_split_selected",
  "input_depth": 50,
  "output_depth": 50,
  "rank_origin": 1,
  "tie_break": "score_desc_then_chunk_id_asc",
  "missing_component_policy": "use_available_component_contribution",
  "duplicate_chunk_policy": "one_contribution_per_system",
  "status": "frozen_before_hybrid_results"
}
```

C.5 Prompt-RAG Configuration

```
{
  "system": "prompt_rag_claude",
  "model": "claude-haiku-4-5-20251001",
  "temperature": 0,
  "max_output_tokens": 4096,
  "candidate_source": "bm25_top25_union_faiss_top25",
  "union_recall_at_25": 0.880,
  "score_range": [0, 3],
  "output_format": "json_schema",
  "tie_break": "score_desc_then_chunk_id_asc",
  "cost_cap_usd": 2.5,
  "status": "frozen_before_results"
}
```

C.6 Evaluation Protocol Configuration

```
{
  "primary_metric": "mrr_at_10",
  "secondary_metrics": ["ndcg_at_10", "recall_at_10",
    "precision_at_5", "precision_at_10", "hit_rate"],
  "r4_development_queries": 60,
  "r4_locked_test_queries": 40,
  "holdout_queries": 12,
  "h5_panel_answers": 150,
  "h5_min_faithfulness_sd": 0.10,
  "bootstrap_resamples": 10000,
  "multiplicity_correction": "holm",
  "alpha": 0.05,
  "h1_equivalence_margin": 0.05,
  "retry_policy": "zero_retry",
  "config_freeze": "enforced_pre_holdout"
}
```

APPENDIX D: WORKED BENCHMARK EXAMPLES

This appendix illustrates the benchmark at the level of individual questions, so that the aggregate metrics in Section 5 can be connected to concrete retrieval behaviour. Examples are drawn from the R4 benchmark over 130 documents and 954 section-aware chunks.

D.1 Representative Queries by Category

Table D.1: Example queries from each benchmark category

Category	Representative query	Tests
Exact match	What is the Pradhan Mantri Mudra Yojana?	Direct scheme identification
Terminology-heavy	Which schemes support self-help group members?	Domain vocabulary matching
Paraphrase	Who can open a zero-balance bank account?	Vocabulary mismatch handling
Entity relation	Which ministry administers PM-KISAN?	Entity-to-entity linkage
Multi-hop	Which scheme helps farmers obtain equipment loans, and through which institution?	Chained reasoning
Synthesis	What support exists for women entrepreneurs?	Cross-document composition

D.2 Worked Example: RRF Fusion Arithmetic

The following trace shows fusion for a single query, reproducing the audit record emitted by the hybrid layer. It demonstrates both the score computation and the deterministic tie-break.

Query: "Which ministry administers PM-KISAN?"

chunk_id	bm25_rank	graph_rank	bm25_rrf	graph_rrf	total	final
chunk-041	1	5	0.016393	0.015385	0.031778	1
chunk-017	5	1	0.015385	0.016393	0.031778	2 <- tie, id asc
chunk-093	2	-	0.016129	0.000000	0.016129	3
chunk-008	-	2	0.000000	0.016129	0.016129	4 <- tie, id asc

Note: chunk-041 and chunk-017 receive identical totals because their component ranks are mirrored. The tie-break orders by chunk_id ascending, making the output reproducible across runs.

D.3 Worked Example: MRR Computation

The metric definitions in Section 3.3 are applied here to a single query so that the aggregate figures can be traced to their inputs.

```
Query: "Who can open a zero-balance bank account?"
Gold (relevant) chunks: {chunk-052, chunk-055}
```

```
BM25 ranking: [chunk-052, chunk-019, chunk-055, chunk-007, ...]
First relevant at rank 1 -> reciprocal rank = 1/1 = 1.000
Recall@10 = 2 gold retrieved / 2 gold total = 1.000
```

```
FAISS ranking: [chunk-019, chunk-007, chunk-031, chunk-052, ...]
First relevant at rank 4 -> reciprocal rank = 1/4 = 0.250
Recall@10 = 1 gold retrieved / 2 gold total = 0.500
```

Single-question contributions of this kind aggregate into category means. With two questions per holdout category, one such question moves the mean by half its range, which is why category orderings are not stable.

D.4 Worked Example: Graph Seeding and Fallback

This trace shows the lexical fallback path taken when query entity extraction returns no seed. The v4 retriever adds alias expansion and dual-entity coverage specifically to reduce how often this path fires.

```
Query: "Which scheme helps farmers obtain equipment loans,
       and through which institution?"
```

```
Step 1 Entity extraction on query text
       pattern: paired_long_form_acronym -> no match
       pattern: law_or_policy           -> no match
       pattern: controlled_acronym      -> no match
       registry lookup: "farmers", "equipment loans", "institution"
                               -> no canonical entity
```

```
Step 2 Seed set is empty -> graph traversal not attempted
Step 3 Fallback engaged -> results served by BM25
```

Outcome: for this query the graph condition is, in effect, the lexical condition. Fallback firing is logged per query, so the share of results genuinely produced by traversal is recoverable from the run artefacts.

D.5 Relevance Judgment Distribution

Table D.2: Relevance judgment provenance

Tier	Judgments	Provenance
R4 benchmark	140 crosswalk rows	Owner-reviewed mapping onto R4 chunks

Tier	Judgments	Provenance
R4 benchmark	1 intentional grade 0	Retained deliberately, disclosed
Independent holdout	21 judgments	Owner-reviewed, authored post-freeze
Synthesis candidates	20 questions	Automated, diagnostic only

The R4 judgments derive from a crosswalk mapping judgments made on an earlier segmentation onto the rebuilt 954-chunk corpus. The owner reviewed every row, but a crosswalk is a weaker instrument than fresh judging on the new chunks, and minimum frozen-reference coverage of 0.5111 indicates that some mappings are partial. This limitation is carried into Section 6.3.